T-PATCHGNN: Efficient Forecasting for Irregular and Asynchronous Time Series

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Eliminating pre-alignment while capturing dynamic correlations with Transformers and GNNs

Introduction

About

This paper proposes **Transformable Patching Graph Neural Networks (T-PATCHGNN)**, a novel framework for Irregular Multivariate Time Series (IMTS) forecasting that avoids canonical pre-alignment while effectively capturing both **intra-time series dependencies and inter-time series correlations**. The model integrates **Transformer-based sequential modeling** and **time-adaptive graph neural networks** to capture dynamic inter-

variable relationships. Extensive experiments on diverse real-world datasets demonstrate that T-PATCHGNN achieves **state-of-the-art forecasting accuracy, superior scalability, and significantly improved efficiency** compared to existing approaches.

Background

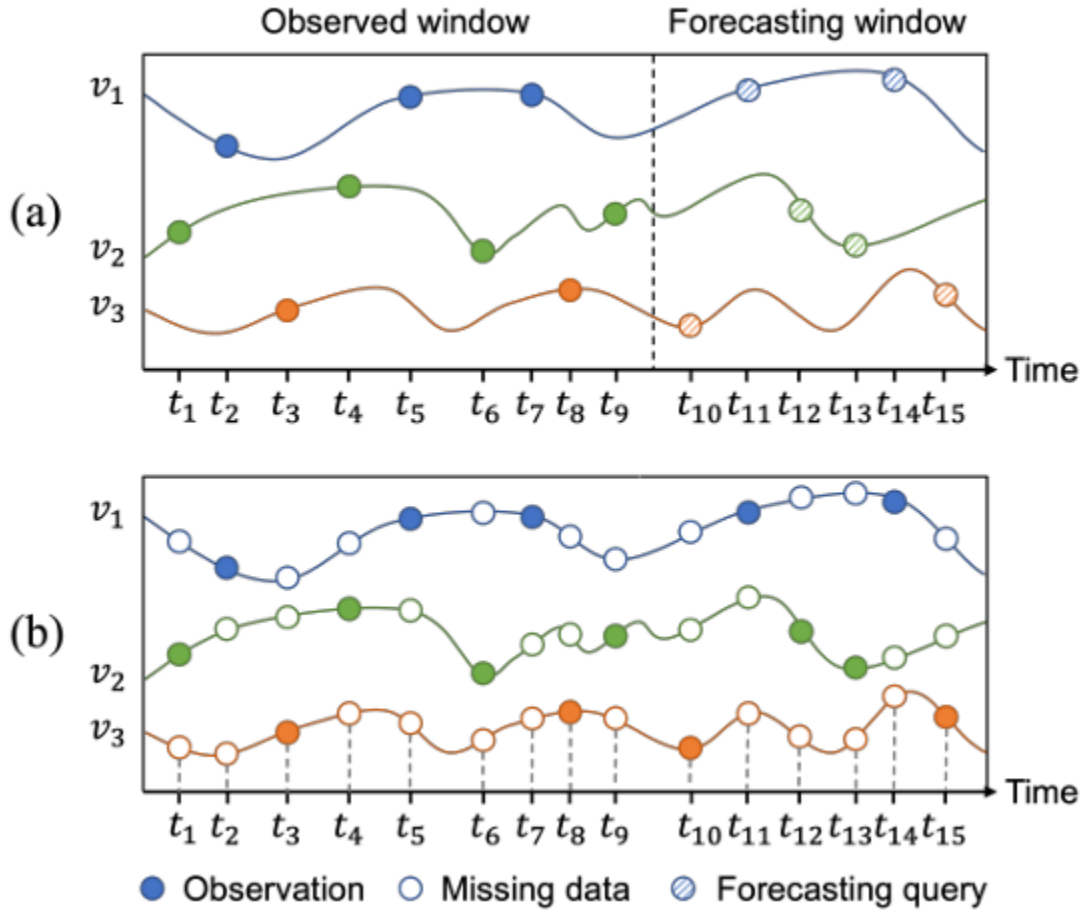

Fig 1. (a) Irregular multivariate time series forecasting problem, where v1, v2, and v3 represent three different vari ables. (b) Canonical pre-alignment representation causes the average sequence length to increase from 5 to 15, an explosive growth proportional to the variable count. (Source)

An **irregular multivariate time series (IMTS)** consists of multiple time-dependent variables, each observed at **non-uniform and asynchronous timestamps**. Formally, different variables may

have distinct observation times and frequencies, leading to severe misalignment across dimensions. As illustrated in **Figure 1** of the paper, even when variables cover the same overall time span, **their observations may be sparse**, **unevenly distributed**, and **temporally inconsistent**. This makes it difficult to directly apply standard multivariate forecasting models, which typically assume synchronized inputs.

A common approach to handling IMTS is to convert them into **canonical pre-aligned representations** by enumerating all unique timestamps across variables and filling missing values with masks. Although this representation simplifies downstream modeling, it introduces a critical drawback: **the resulting sequence length grows rapidly with the number of variables**, causing a **sequence length explosion** that severely limits scalability. Alternative continuous-time models, such as neural ODE–based methods, address irregularity by modeling latent state evolution between observations, but they often incur **high computational cost** and focus primarily on intra-variable dynamics. These limitations motivate the need for a representation that can simultaneously handle **irregularity, asynchrony, and inter-variable dependencies** without sacrificing efficiency.

Preliminary

Problem Definition

The paper formulates Irregular Multivariate Time Series (IMTS) as **a collection of observations across multiple variables**, where each variable is sampled at its own irregular timestamps. Formally, an IMTS is defined as:

$$O = \left\{ o^n_{1:L_n} \right\}_{n=1}^{N} = \left\{ \left[(t^n_i, x^n_i) \right]_{i=1}^{L_n} \right\}_{n=1}^{N}$$

where N denotes the number of variables, and each variable n contains, L_n observations consisting of timestamp–value pairs. Given historical observations and a set of forecasting queries $Q = \{q_j^n\}$,the objective is to learn a forecasting function:

$$F(\mathcal{O}, \mathcal{Q}) \rightarrow \hat{X}$$

that **predicts future values corresponding to the query timestamps.**

Canonical Pre-Alignment Representation for IMTS

A common approach to facilitate modeling is the canonical pre-alignment representation, which constructs a unified timeline T from all unique timestamps and represents the data using three matrices: timestamps T, values X, and masking matrix M. While this representation enables compatibility with standard sequence models, it introduces a critical scalability issue. Specifically, the resulting sequence length L depends on the total number of unique timestamps across all variables and satisfies:

$$L_{\max} \leq L \leq N \cdot L_{\text{avg}}$$

where L_{av9} and L_{max} denote the average and maximum number of observations per variable, respectively.

This implies that **the sequence length can grow proportionally with the number of variables**, leading to substantial computational and memory overhead. This phenomenon, known as *sequence length explosion*, **highlights the limitations of pre-alignment strategies** and **motivates alternative representations for IMTS forecasting.**

Method

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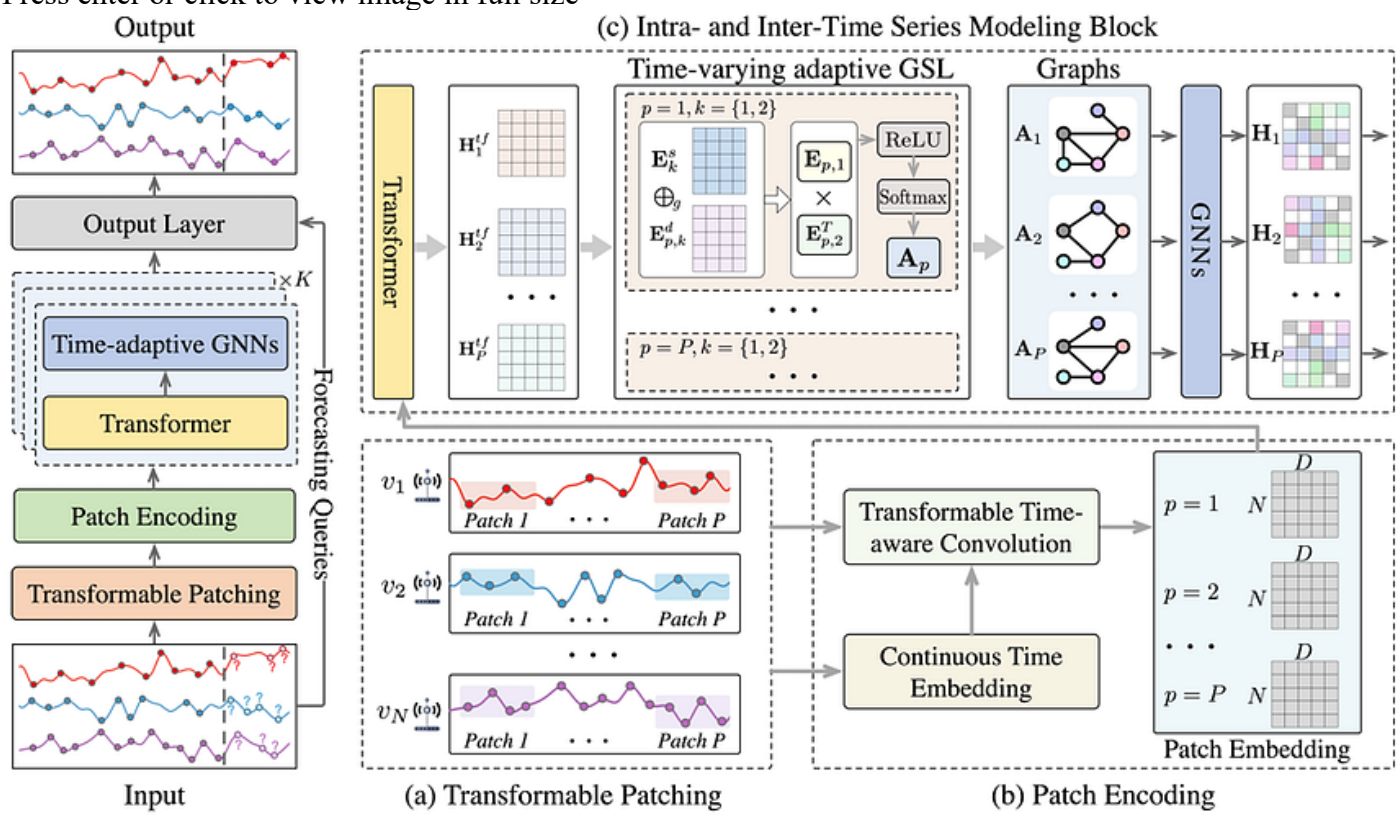

Fig 2. Overview of T-PATCHGNN, which initially divides each univariate irregular time series into a series of trans formable patches with varying number of consecutive observations but maintains a unified time horizon resolution. (Source)

The authors in this literature introduces **Transformable Patching Graph Neural Networks (T-PATCHGNN)** for irregular multivariate time series forecasting. As illustrated in **Figure 2**, the framework consists of four major components: **irregular time series patching**, **patch encoding**, **intra-time series modeling**, and **inter-time series modeling**, followed by a forecasting head. The key design principle is to reformulate IMTS forecasting at a **patch-level temporal resolution**, which naturally aligns asynchronous observations while avoiding canonical pre-alignment and its associated scalability issues.

Irregular Time Series Patching

Transformable Patching

The core idea of transformable patching is to divide each univariate irregular time series into a sequence of patches with **variable numbers of observations but fixed temporal span**. Unlike standard patching, which groups a fixed number of points, this method ensures that each patch corresponds to a consistent time horizon (*e.g.*, 2 hours), thereby normalizing temporal resolution across variables. **This design resolves the inconsistency caused by irregular sampling, where equal-sized observation windows may span drastically different time intervals.**

Importantly, **patching is performed independently for each variable,** avoiding the need for global time alignment. As a result, **the**

sequence length depends only on the number of patches per variable rather than the total number of unique timestamps across variables. This directly mitigates the sequence length explosion problem described in the preliminary section. Moreover, **the patch-level representation provides richer local context**, which improves the model's ability to capture short-term temporal dynamics.

Patch Encoding

Once patches are constructed, each patch is encoded into a latent representation. The model first introduces a **continuous time embedding** to represent irregular timestamps:

$$\phi(t)[d] = \begin{cases} \omega_0 t + \alpha_0, & d = 0 \\ \sin(\omega_d t + \alpha_d), & 0 < d < D_t \end{cases}$$

This formulation combines linear and periodic components, allowing the model to capture both non-periodic trends and periodic patterns in time series data.

The embedded observations within a patch are then processed using a **Transformable Time-aware Convolution Network (TTCN)**, which dynamically generates convolutional filters conditioned on the input sequence:

$$f_d = \left[\frac{\exp(F_d(z_i))}{\sum_{j=1}^{L_p} \exp(F_d(z_j))} \right]_{i=1}^{L_p}$$

where the filter adapts to variable-length inputs. The resulting patch embedding is computed via:

$$h_p^c = \left[\sum_{i=1}^{L_p} f_d[i]^\top z_i \right]_{d=1}^{D-1}$$

This design is particularly suitable for IMTS because **it can handle variable-length patches and irregular intervals without introducing additional parameters.** A masking term is further incorporated to account for empty patches:

$$h_p = [h_p^c \| m_p]$$

ensuring robustness under sparse observations.

Intra- and Inter-Time Series Modeling

Transformer for Intra-Time Series Modeling

After encoding, patch embeddings are treated as tokens and fed into a Transformer to model intra-series dependencies. Positional encodings are added to preserve temporal ordering, and multi-head self-attention is applied:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V$$

This allows **the model to capture long-range dependencies across patches**, effectively modeling temporal dynamics within each variable.

Time-Varying Adaptive Graph Learning

To model inter-series relationships, the authors propose learning **time-varying adaptive graphs**. Each graph corresponds to a patch-level time step and is constructed by combining static variable embeddings with dynamic patch representations:

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$$E_{p,k} = E_k^s + g_{p,k} \cdot E_{p,k}^d$$

where the gating mechanism:

$$g_{p,k} = \mathrm{ReLU}(\tanh([H_p^{tf} \| E_k^s] W_k^g))$$

controls the contribution of dynamic information.

The adjacency matrix for each patch is then computed as:

$$A_p = \mathrm{Softmax}(\mathrm{ReLU}(E_{p,1} E_{p,2}^\top))$$

This formulation enables the model to capture **dynamic and context-dependent correlations** between variables, which is essential for IMTS where relationships evolve over time.

GNN for Inter-Time Series Modeling

Given the learned graphs, Graph Neural Networks are applied to propagate information across variables:

$$H_p = \mathrm{ReLU}\left(\sum_{m=0}^{M} (A_p)^m H_p^{tf} W_m^{gnn}\right)$$

This operation aggregates multi-hop relational information, allowing **the model to capture both direct and indirect dependencies between variables.** By stacking multiple such blocks, the framework can flexibly model complex temporal and relational patterns.

IMTS Forecasting

After intra- and inter-series modeling, the patch-level representations are flattened and projected into a latent space:

$$H = \text{Flatten}([H_p])W_f$$

For each forecasting query q_j^n, the model concatenates the latent representation with its time embedding and predicts the output via an MLP:

$$\hat{x}_j^n = \text{MLP}([H_n \| \phi(q_j^n)])$$

The model is trained using Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \frac{1}{Q_n} \sum_{j=1}^{Q_n} (\hat{x}_j^n - x_j^n)^2$$

Scalability Analysis

The proposed transformable patching ensures that the effective sequence length is proportional to the average number of observations per variable:

$$L_{tp} = L_{avg} \leq L_{cpr} \leq N \cdot L_{avg}$$

This avoids the exponential growth seen in pre-alignment approaches. The canonical alignment can amplify sequence length by up to 20× or more, whereas the proposed method maintains a stable and scalable representation. **This property is critical for handling high-dimensional IMTS data efficiently**.

Experiment

The experiment in this literature aimed to comprehensively evaluate the effectiveness of T-PATCHGNN on **irregular multivariate time series forecasting tasks**. The authors constructed a diverse benchmark covering multiple real-world domains and compared against a wide range of baselines spanning *forecasting*, *classification*, and *interpolation* tasks. Through these experiments, they assess not only predictive performance but also scalability and robustness of the proposed method.

Dataset

The evaluation is conducted on four real-world IMTS datasets spanning different domains:

- **PhysioNet:** Healthcare dataset with physiological signals (high irregularity and sparsity)
- **MIMIC:** Clinical dataset with intensive care unit measurements

- **Human Activity:** Biomechanics dataset with sensor-based motion data
- **USHCN:** Climate dataset with long-term environmental observations

These datasets vary significantly in terms of number of variables, sampling frequency, and sparsity, providing a comprehensive testbed for IMTS forecasting.

Model Set-up

The experiments are conducted under consistent settings to ensure fairness:

- **Hidden dimensions**: 64 (PhysioNet, MIMIC), 32 (others)
- **Batch size**: 32 (most datasets), 192 (USHCN)
- **Optimizer**: Adam
- **Early stopping**: patience of 10 epochs
- Multiple runs with different random seeds

Key configurations of T-PATCHGNN include:

- **Patch window sizes tailored per dataset** (*e.g.*, 8 hours, 300 ms, 2 months)

- **Transformer heads**: 1
- **GNN layers**: 1
- **Number of blocks:** 1
- **Learning rate:** 0.001

Comparison

The authors compare T-PATCHGNN against 17 baselines across four categories:

- **MTS Forecasting Models:**
  DLinear, TimesNet, PatchTST, Crossformer, Graph WaveNet, MTGNN, StemGNN, CrossGNN, FourierGNN
- **IMTS Classification Models:**
  GRU-D, SeFT, RainDrop, Warpformer
- **IMTS Interpolation Models:**
  mTAND
- **IMTS Forecasting Models:**
  Latent-ODE, CRU, Neural Flow

This broad comparison ensures that the proposed method is evaluated against both specialized IMTS models and general time series architectures.

Evaluation Metrics

Two standard regression metrics are used:

- **Mean Squared Error (MSE):**

$$\mathrm{MSE} = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{Q_n}\sum_{j=1}^{Q_n}(\hat{x}_j^n - x_j^n)^2$$

MSE is sensitive to large errors and emphasizes robustness to outliers.

- **Mean Absolute Error (MAE):**

$$\mathrm{MAE} = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{Q_n}\sum_{j=1}^{Q_n}|\hat{x}_j^n - x_j^n|$$

MAE provides a more interpretable measure of average prediction error.

These complementary metrics provide a balanced evaluation of forecasting accuracy. Note that these metrics stand for **multivariate forecasting evaluation**.

Results

Main Results

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Algorithm	PhysioNet		MIMIC		Human Activity		USHCN	
	MSE×10^{-3}	MAE×10^{-2}	MSE×10^{-2}	MAE×10^{-2}	MSE×10^{-3}	MAE×10^{-2}	MSE×10^{-1}	MAE×10^{-1}
DLinear	41.86 ± 0.05	15.52 ± 0.03	4.90 ± 0.00	16.29 ± 0.05	4.03 ± 0.01	4.21 ± 0.01	6.21 ± 0.00	3.88 ± 0.02
TimesNet	16.48 ± 0.11	6.14 ± 0.03	5.88 ± 0.08	13.62 ± 0.07	3.12 ± 0.01	3.56 ± 0.02	5.58 ± 0.05	3.60 ± 0.04
PatchTST	12.00 ± 0.23	6.02 ± 0.14	3.78 ± 0.03	12.43 ± 0.10	4.29 ± 0.14	4.80 ± 0.09	5.75 ± 0.01	3.57 ± 0.02
Crossformer	6.66 ± 0.11	4.81 ± 0.11	2.65 ± 0.10	9.56 ± 0.29	4.29 ± 0.20	4.89 ± 0.17	5.25 ± 0.04	3.27 ± 0.09
Graph Wavenet	6.04 ± 0.28	4.41 ± 0.11	2.93 ± 0.09	10.50 ± 0.15	2.89 ± 0.03	3.40 ± 0.05	5.29 ± 0.04	3.16 ± 0.09
MTGNN	6.26 ± 0.18	4.46 ± 0.07	2.71 ± 0.23	9.55 ± 0.65	3.03 ± 0.03	3.53 ± 0.03	5.39 ± 0.05	3.34 ± 0.02
StemGNN	6.86 ± 0.28	4.76 ± 0.19	1.73 ± 0.02	7.71 ± 0.11	8.81 ± 0.37	6.90 ± 0.02	5.75 ± 0.09	3.40 ± 0.09
CrossGNN	7.22 ± 0.36	4.96 ± 0.12	2.95 ± 0.16	10.82 ± 0.21	3.03 ± 0.10	3.48 ± 0.08	5.66 ± 0.04	3.53 ± 0.05
FourierGNN	6.84 ± 0.35	4.65 ± 0.12	2.55 ± 0.03	10.22 ± 0.08	2.99 ± 0.02	3.42 ± 0.02	5.82 ± 0.06	3.62 ± 0.07
GRU-D	5.59 ± 0.09	4.08 ± 0.05	1.76 ± 0.03	7.53 ± 0.09	2.94 ± 0.05	3.53 ± 0.06	5.54 ± 0.38	3.40 ± 0.28
SeFT	9.22 ± 0.18	5.40 ± 0.08	1.87 ± 0.01	7.84 ± 0.08	12.20 ± 0.17	8.43 ± 0.07	5.80 ± 0.19	3.70 ± 0.11
RainDrop	9.82 ± 0.08	5.57 ± 0.06	1.99 ± 0.03	8.27 ± 0.07	14.92 ± 0.14	9.45 ± 0.05	5.78 ± 0.22	3.67 ± 0.17
Warpformer	5.94 ± 0.35	4.21 ± 0.12	1.73 ± 0.04	7.58 ± 0.13	2.79 ± 0.04	3.39 ± 0.03	5.25 ± 0.05	3.23 ± 0.05
mTAND	6.23 ± 0.24	4.51 ± 0.17	1.85 ± 0.06	7.73 ± 0.13	3.22 ± 0.07	3.81 ± 0.07	5.33 ± 0.05	3.26 ± 0.10
Latent-ODE	6.05 ± 0.57	4.23 ± 0.26	1.89 ± 0.19	8.11 ± 0.52	3.34 ± 0.11	3.94 ± 0.12	5.62 ± 0.03	3.60 ± 0.12
CRU	8.56 ± 0.26	5.16 ± 0.09	1.97 ± 0.02	7.93 ± 0.19	6.97 ± 0.78	6.30 ± 0.47	6.09 ± 0.17	3.54 ± 0.18
Neural Flow	7.20 ± 0.07	4.67 ± 0.04	1.87 ± 0.05	8.03 ± 0.19	4.05 ± 0.13	4.46 ± 0.09	5.35 ± 0.05	3.25 ± 0.05
T-PATCHGNN	$\mathbf{4.98 \pm 0.08}$	$\mathbf{3.72 \pm 0.03}$	$\mathbf{1.69 \pm 0.03}$	$\mathbf{7.22 \pm 0.09}$	$\mathbf{2.66 \pm 0.03}$	$\mathbf{3.15 \pm 0.02}$	$\mathbf{5.00 \pm 0.04}$	$\mathbf{3.08 \pm 0.04}$

Table 1: Overall performance evaluated by MSE and MAE. Here, the best-performing and second-best results are highlighted in bold and underline, respectively. (Source)

The main experimental results, reported in **Table 1**, demonstrate that T-PATCHGNN consistently achieves the best performance across all four datasets in terms of both MSE and MAE. Notably, the proposed method outperforms the second-best baseline by a significant margin, often exceeding 10% improvement. For instance, on the PhysioNet dataset, T-PATCHGNN achieves an MSE of 4.98×10^{-3}, surpassing strong baselines such as GRU-D and Warpformer. Similar trends are observed on *MIMIC*, H*uman Activity*, and *USHCN* datasets, indicating robust generalization across domains.

A key observation is that conventional MTS forecasting models, including Transformer-based and GNN-based approaches (e.g., PatchTST, MTGNN), fail to maintain competitive performance when directly applied to IMTS. This highlights the intrinsic difficulty of irregular and asynchronous data. Furthermore, existing IMTS-specific

models, particularly those based on neural ODEs, also underperform compared to T-PATCHGNN, **suggesting that solely modeling continuous-time dynamics is insufficient without explicitly capturing inter-series correlations**. These results **validate the effectiveness of combining transformable patching with adaptive graph learning.**

Ablation Study

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Ablation	PhysioNet		MIMIC		Human Activity		USHCN	
	MSE$\times 10^{-3}$	MAE$\times 10^{-2}$	MSE$\times 10^{-2}$	MAE$\times 10^{-2}$	MSE$\times 10^{-3}$	MAE$\times 10^{-2}$	MSE$\times 10^{-1}$	MAE$\times 10^{-1}$
Complete	$\mathbf{4.98 \pm 0.08}$	$\mathbf{3.72 \pm 0.03}$	$\mathbf{1.69 \pm 0.03}$	$\mathbf{7.22 \pm 0.09}$	$\mathbf{2.66 \pm 0.03}$	$\mathbf{3.15 \pm 0.02}$	$\mathbf{5.00 \pm 0.04}$	$\mathbf{3.08 \pm 0.04}$
w/o Patch	5.27 ± 0.06	3.88 ± 0.03	1.73 ± 0.02	7.41 ± 0.06	2.85 ± 0.05	3.28 ± 0.02	5.32 ± 0.02	3.30 ± 0.09
rp Patch	5.64 ± 0.13	3.97 ± 0.05	1.72 ± 0.01	7.29 ± 0.02	2.82 ± 0.02	3.29 ± 0.04	5.20 ± 0.07	3.17 ± 0.07
w/o VE	10.53 ± 0.15	5.30 ± 0.01	3.71 ± 0.02	12.06 ± 0.03	2.79 ± 0.03	3.18 ± 0.03	5.08 ± 0.03	3.25 ± 0.08
w/o PE	5.10 ± 0.02	3.83 ± 0.06	1.71 ± 0.01	7.42 ± 0.11	2.79 ± 0.02	3.21 ± 0.04	5.08 ± 0.02	3.22 ± 0.08
w/o Transformer	5.10 ± 0.12	3.83 ± 0.07	1.71 ± 0.02	7.27 ± 0.06	2.78 ± 0.04	3.20 ± 0.04	5.22 ± 0.06	3.19 ± 0.11

Table 2. Ablation results of T-PATCHGNN on four datasets. (Source)

The ablation results, summarized in **Table 2**, evaluate the contribution of each component in T-PATCHGNN. Removing transformable patching (w/o Patch) leads to noticeable performance degradation across all datasets, confirming its critical role in handling irregularity and enabling effective downstream modeling. Replacing transformable patching with standard patching (rp Patch) further degrades performance in some cases, demonstrating that maintaining a consistent temporal resolution is essential for IMTS.

The study also **highlights the importance of modeling inter-series correlations**. Removing variable embeddings (w/o VE) results in a substantial drop in performance, particularly in healthcare

datasets such as *PhysioNet* and *MIMIC*, where inherent relationships between variables are crucial. In contrast, removing dynamic patch embeddings (w/o PE) shows relatively smaller degradation, **suggesting that static variable characteristics may play a more dominant role in certain domains.**

Additionally, removing the Transformer module leads to consistent but moderate performance decline, indicating that **intra-series temporal modeling remains important but is less critical than proper representation and correlation modeling**. Overall, the ablation study confirms that e**ach component contributes meaningfully to the overall performance**, **with transformable patching and graph-based correlation modeling being the most impactful.**

Scalability and Efficiency Analysis

Description	PhysioNet	MIMIC	Human Activity	USHCN
# Variable	41	96	12	5
Avg # observations	10.7	1.5	30.2	36.1
Avg aligned length	75	46.1	120.6	163.9
Max aligned length	216	643	130	214
Avg amplification	$\times 7.0$	$\times 21.9$	$\times 4.0$	$\times 4.5$
Max amplification	$\times 16.8$	$\times 96.0$	$\times 4.0$	$\times 5.0$

Table 3. Evidence of sequence length explosion. Aligned length represents the sequence length after canonical prealignment. (Source)

The scalability analysis, presented in **Table 3**, **provides empirical evidence of the sequence length explosion problem in canonical pre-alignment representations**. For example, in

the *MIMIC* dataset, the average sequence length increases by approximately 21.9× compared to the original number of observations, with a maximum amplification of up to 96×. Similar trends are observed across other datasets, especially as the number of variables increases.

In contrast, T-PATCHGNN avoids this issue by operating directly on patch-level representations without global alignment. As a result, t**he effective sequence length remains proportional to the average number of observations per variable, significantly improving scalability.**

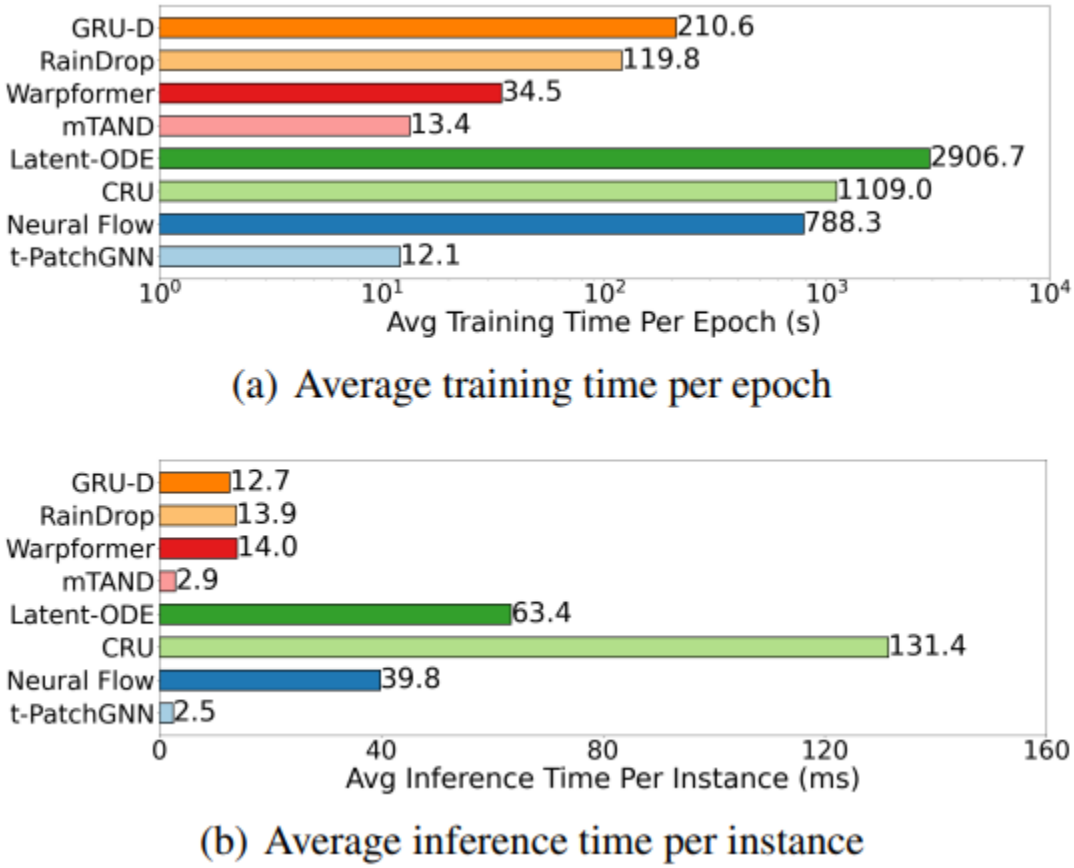

Fig 3. Efficiency comparison of training and inference. (Source)

Furthermore, **Figure 3** compares training and inference efficiency. The proposed method demonstrates **competitive or superior efficiency compared to baseline methods**, particularly in **scenarios with high dimensionality**. This confirms that the proposed approach not only improves predictive performance but also **offers practical advantages in terms of computational cost and scalability**.

Effect of Patch Size

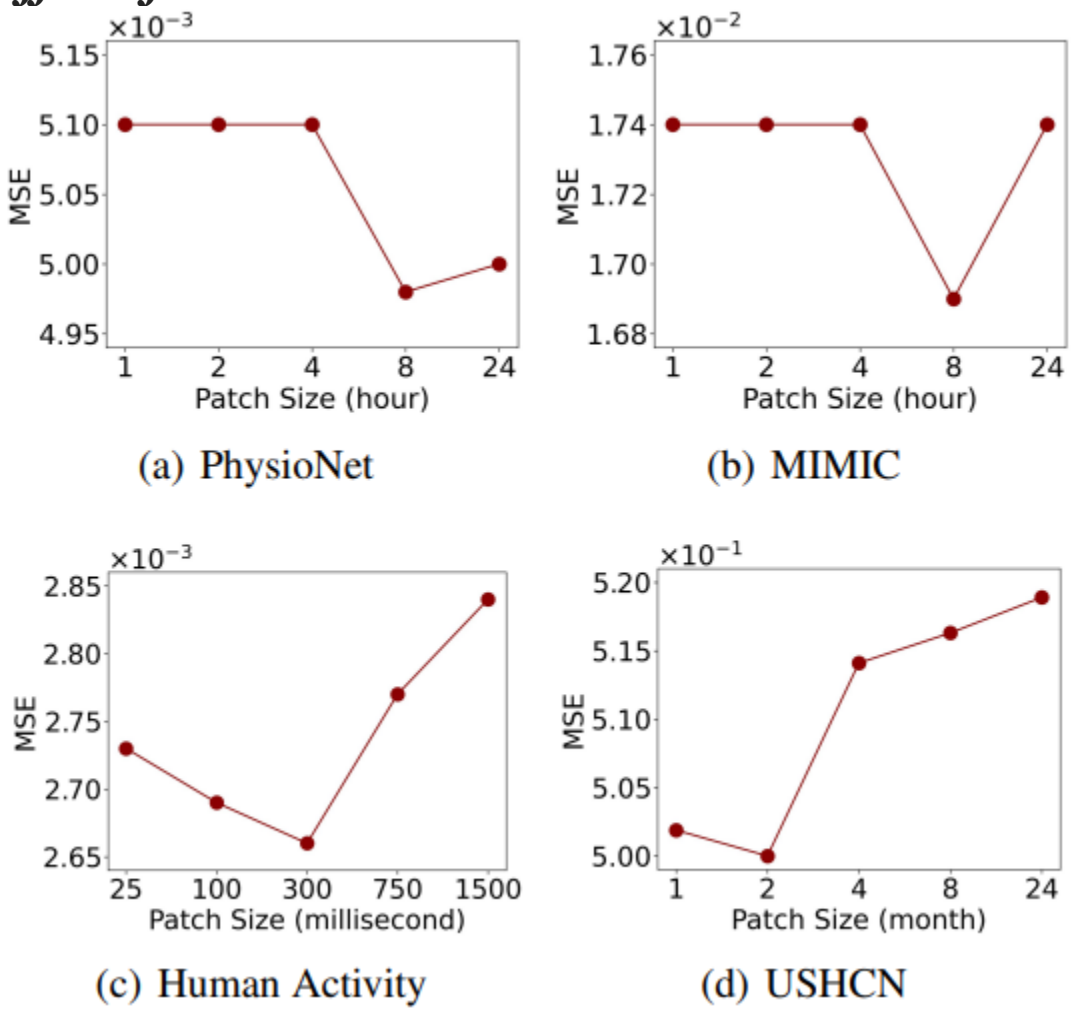

Fig 4. Effect of different patch sizes. (Source)

This experiment analyzes the impact of patch window size on forecasting performance, as illustrated in **Figure 4**. The results show that optimal patch size is dataset-dependent.
For **PhysioNet** and **MIMIC**, performance is stable for small patches and peaks around **8 hours**, balancing sufficient observations within each patch and temporal resolution. Too small patches lack informative context, while overly large patches degrade resolution.

In contrast, **Human Activity** and **USHCN** benefit from **smaller patch sizes**, as their more dynamic patterns require finer temporal granularity. The authors further note that **larger patches favor long-range forecasting** by capturing broader trends, whereas **smaller patches are better for short-range predictions**. This highlights **patch size as a key factor controlling the trade-off between temporal resolution and contextual richness.**

Conclusion

This paper introduces **T-PATCHGNN**, a novel framework for irregular multivariate time series forecasting that effectively addresses the challenges of irregular sampling, asynchrony, and scalability. By proposing **transformable patching**, the method **avoids the limitations of canonical pre-alignment while preserving local temporal semantics**. The integration of Transformer-based intra-series modeling and time-adaptive graph neural networks enables comprehensive modeling of both temporal dependencies and dynamic

inter-variable correlations. Extensive experiments on diverse real-world datasets demonstrate **consistent performance improvements over a wide range of baselines.**

Review

This paper presents a thoughtfully designed framework that tackles three fundamental challenges in IMTS forecasting: *irregularity*, *asynchrony*, and *scalability*. The key strength lies in **the transformable patching mechanism**, which elegantly **replaces the widely used pre-alignment strategy and directly addresses the sequence length explosion issue**. By combining this representation with Transformer-based temporal modeling and adaptive graph learning, **the model captures both intra- and inter-series dependencies in a unified manner.** The experimental evaluation is comprehensive, covering multiple domains and a wide range of baselines, and the consistent performance gains strongly support the effectiveness of the proposed approach. **The inclusion of scalability analysis further strengthens the practical relevance of the work.**

However, the framework also introduces a relatively **complex architecture that integrates multiple components**, which may **increase implementation difficulty and hyperparameter sensitivity**. While the model demonstrates strong empirical performance, its **reliance on carefully designed patch sizes and graph learning mechanisms** may limit its robustness in highly

heterogeneous settings. Also, although the method avoids ODE-based computation, **the combination of Transformer and GNN modules could still pose challenges for deployment in resource-constrained environments.** Future work could explore simplifying the architecture, automating patch construction, or incorporating more lightweight modeling techniques. Despite these considerations, **the paper makes a significant contribution to IMTS forecasting and provides a promising direction for future research in handling irregular and asynchronous time series data.**

Reference

Irregular Multivariate Time Series Forecasting: A Transformable Patching Graph Neural Networks Approach